



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105



CSA0822-Python Programming(Slot B)

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Python Data Type Program Questions

1. Integer & Float

Question:

Write a program that takes the principal amount, rate of interest, and time period as input and calculates simple interest. Use integer and float data types appropriately.

```
File Edit Format Run Options Window Help
principal=float(input("enter principal"))
rate=float(input("enter rate"))
time=int(input("enter time"))
si=(principal*rate*time)/100
total=principal+si
print("simple interest:",si)
print("total amount",total)
```

output

```
IDLE Shell 3.14.8
File Edit Shell Debug Options Window Help
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
enter principal40
enter rate23
enter time20
simple interest: 184.0
total amount 224.0
>>>
```

2. String

Question:

Write a program that accepts a full name from the user and prints:

- Number of characters
- Name in uppercase and lowercase
- First and last name separately

```
File Edit Format Run Options Window Help
name=input("enter name")
characters=len(name)
uppercase=name.upper()
lowercase=name.lower()
nameparts=name.split()
firstname=nameparts[0]
lastname=nameparts[-1]
print("name:",name)
print("no of characters:",characters)
print("uppercase:",uppercase)
print("lowercase:",lowercase)
print("first name",firstname)
print("last name",lastname)
```

output

```
IDLE Shell 3.14.8
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Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
enter name:tharani rb
name: tharani rb
no of characters: 10
uppercase: THARANI RB
lowercase: tharani rb
first name tharani
last name rb
>>>
```

3. Boolean

Question:

Write a program that checks whether a student has passed or failed based on marks (pass mark ≥ 40). Use Boolean data type to store the result.

data types py.py - C:/Users/thara/OneDrive/Desktop/data types py.py (3.14.8)

```
File Edit Format Run Options Window Help
marks=float(input("enter marks:"))
passmark=marks>=40
if passmark:
    print("result:pass")
else:
    print("result:fail")
|
```

output

```
DLE Shell 3.14.8
Edit Shell Debug Options Window Help
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
enter marks:56
result:pass

===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py ==
enter marks:36
result:fail
```

4. List

Question:

Create a program that stores marks of 5 subjects in a list, calculates total and average, and finds the highest and lowest marks.

```
data types py.py - C:/Users/thara/OneDrive/Desktop/data types py.py (3.14.8)
File Edit Format Run Options Window Help

marks = []
print("Enter the marks for 5 subjects:")
for i in range(1, 6):
    while True:
        try:
            # Read input and convert to float for fractional marks
            score = float(input(f"Subject {i}: "))
            if 0 <= score <= 100: # Validates standard mark boundaries
                marks.append(score)
                break
            else:
                print("Please enter a valid mark between 0 and 100.")
        except ValueError:
            print("Invalid input. Please enter a numerical value.")
total_marks = sum(marks)
average_marks = total_marks / len(marks)
highest_mark = max(marks)
lowest_mark = min(marks)
print("\n--- Performance Summary ---")
print(f"All Marks:      {marks}")
print(f"Total Marks:     {total_marks:.2f}")
print(f"Average Marks:    {average_marks:.2f}")
print(f"Highest Mark:     {highest_mark:.2f}")
print(f"Lowest Mark:      {lowest_mark:.2f}")
```

output

```
*IDLE Shell 3.14.8*
File Edit Shell Debug Options Window Help

Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 b
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
Enter the marks for 5 subjects:
Subject 1: 67
Subject 2: 78
Subject 3: 98
Subject 4: 78
Subject 5: 65

--- Performance Summary ---
All Marks:      [67.0, 78.0, 98.0, 78.0, 65.0]
Total Marks:    386.00
Average Marks:  77.20
Highest Mark:   98.00
Lowest Mark:    65.00
>>> |
```

5. Tuple

Question:

Store employee details (Employee ID, Name, Department, Salary) in a tuple and display them. Try to modify one value and observe the error.

data types py.py - C:/Users/thara/OneDrive/Desktop/data types py.py (3.14.8)

```
File Edit Format Run Options Window Help
employee_details = (101, "priya", "cse", 85000)
print("--- Employee Details ---")
print("Employee ID:", employee_details[0])
print("name", employee_details[1])
print("branch", employee_details[2])
print("salary", employee_details[3])
try:
    print("\nAttempting to modify the salary...")
    employee_details[3] = 90000
except TypeError as error:
    print(f"Captured Error: {error}")
```

output

```
IDLE Shell 3.14.8
File Edit Shell Debug Options Window Help
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
--- Employee Details ---
Employee ID: 101
name priya
branch cse
salary 85000

Attempting to modify the salary...
Captured Error: 'tuple' object does not support item assignment
>>>
```

6. Dictionary

Question:

Create a dictionary to store student roll numbers as keys and their names as values.

Allow the user to:

- Search a student by roll number
- Add a new student
- Display all students

```
File Edit Format Run Options Window Help
students = {}
while True:
    print("\n--- Student Management Menu ---")
    print("1. Add a new student")
    print("2. Search student by roll number")
    print("3. Display all students")
    print("4. Exit")
    choice = input("Enter your choice (1-4): ")
    if choice == '1':
        roll_no = input("Enter roll number: ")
        name = input("Enter student name: ")
        students[roll_no] = name
        print("Student name added successfully!")
    elif choice == '2':
        roll_no = input("Enter roll number to search: ")
        if roll_no in students:
            print(f"Student Found - Roll No: {roll_no}, Name: {students[roll_no]}")
        else:
            print("Student with this roll number not found.")
    elif choice == '3':
        if not students:
            print("No student records found.")
        else:
            print("\n--- List of All Students ---")
            for roll, name in students.items():
                print(f"Roll No: {roll} | Name: {name}")
    elif choice == '4':
        print("Exiting program. Goodbye!")
        break
    else:
        print("Invalid choice! Please enter a number between 1 and 4.")
```

output

```
File Edit Shell Debug Options Window Help
--- Student Management Menu ---
1. Add a new student
2. Search student by roll number
3. Display all students
4. Exit
Enter your choice (1-4): 1
Enter roll number: 101
Enter student name: levi
Student name added successfully!

--- Student Management Menu ---
1. Add a new student
2. Search student by roll number
3. Display all students
4. Exit
Enter your choice (1-4): 1
Enter roll number: 102
Enter student name: kiki
```

7. Set

Question: Write a program that takes two lists of student names (one from Class A and one from ClassB) and finds:

- Common students
- Students only in Class A
- All unique students

```
data types py.py - C:/Users/thara/OneDrive/Desktop/data types py.py (3.14.8)
File Edit Format Run Options Window Help
def analyze_classes(class_a, class_b):
    set_a = set(class_a)
    set_b = set(class_b)
    common_students = list(set_a.intersection(set_b))
    only_in_a = list(set_a.difference(set_b))
    all_unique_students = list(set_a.union(set_b))
    print(f"• Common students: {common_students}")
    print(f"• Students only in Class A: {only_in_a}")
    print(f"• All unique students: {all_unique_students}")
class_a_students = ["Alice", "Bob", "Charlie", "David"]
class_b_students = ["Charlie", "David", "Edward", "Frank"]
analyze_classes(class_a_students, class_b_students)
```

output

```
IDLE Shell 3.14.8
File Edit Shell Debug Options Window Help
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
• Common students: ['Charlie', 'David']
• Students only in Class A: ['Bob', 'Alice']
• All unique students: ['Alice', 'Bob', 'Charlie', 'Edward', 'Frank', 'David']
>>>
```

8. Mixed Data Types (List of Dictionaries) Question:

Create a list of dictionaries where each dictionary stores product details (name, price, quantity). Calculate the total bill amount.

```
File Edit Format Run Options Window Help
products = [
    {"name": "Apple", "price": 1.5, "quantity": 4},
    {"name": "Banana", "price": 0.5, "quantity": 6},
    {"name": "Milk", "price": 2.0, "quantity": 2}
]
total_bill = 0
for item in products:
    item_cost = item["price"] * item["quantity"]
    total_bill += item_cost

print(f"Total Bill Amount: ${total_bill:.2f}")
```

output

```
IDLE Shell 3.14.8
File Edit Shell Debug Options Window Help
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
Total Bill Amount: $13.00
>>>
```

9. Type Conversion

Question:

Take a number as string input from the user, convert it into integer and float, and perform addition with another number.

```
data types py.py - C:/Users/thara/OneDrive/Desktop/data types py.py (3.14.8)
File Edit Format Run Options Window Help
user_input = input("Enter a number: ")
int_number = int(user_input)
float_number = float(user_input)
another_number = 10
int_result = int_number + another_number
float_result = float_number + another_number
print(f"Integer addition ({int_number} + {another_number}): {int_result}")
print(f"Float addition ({float_number} + {another_number}): {float_result}")
```

output

```
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1400 AMD64] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
Enter a number: 34
Integer addition (34 + 10): 44
Float addition (34.0 + 10): 44.0

>>>
```

10. Nested Data Types

Question:

Create dictionary that stores student name as key and a list of 3 subject marks as value. Calculate average marks for each student.

```
data types py.py - C:/Users/thara/OneDrive/Desktop/data types py.py (v3.14.8)
File Edit Format Run Options Window Help
students = {
    "Arun": [85, 90, 80],
    "Priya": [78, 88, 92],
    "Rahul": [90, 75, 85]
}
for name, marks in students.items():
    average = sum(marks) / len(marks)
    print(name, "Average Marks =", average)
```

output

```
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64-bit AMD64] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/thara/OneDrive/Desktop/data types py.py =====
Arun Average Marks = 85.0
Priya Average Marks = 86.0
Rahul Average Marks = 83.33333333333333

>>>
```